

Deepak Pandidurai

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Professional Summary

PhD Student in Atmospheric science with expertise in extreme weather analysis, climate modeling, and numerical simulation. Proven record of delivering impactful research on anomalous moisture transport and hydrological extremes using WRF, HYSPLIT, and advanced Python-based data analytics.

Education

2024–Pres | **Ph.D. Scholar**, Dept. of Hydrology, IIT Roorkee. (CGPA: 9.57). Focus: Atmospheric Rivers.

2020–22 | **M.Sc. Atmospheric Science**, Central University of Rajasthan. (CGPA: 8.11)

2016–19 | **B.Sc. Physics**, Madras Christian College, University of Madras. (Grade: 74.26%)

Research Experience

PhD Research, IIT Roorkee (2024–Pres): Leading research on atmospheric rivers and moisture transport; managing computational servers and international scientific coordination.

JRF, IISER Mohali (2022–2023): Led climate data retrieval and WRF simulations for ISRO projects; authored publications and automated research workflows.

M.Sc. Thesis (2022): Assessed meteorological drought in Tamil Nadu (1981–2020) using CDO/QGIS and SPI/Aridity Index analysis.

Key Research Skills

Programming: Python, BASH, MATLAB, FORTRAN, CDO, NCL, GrADS.

Modeling: WRF-ARW/DA, HYSPLIT. **Geospatial:** QGIS, ArcMap, GEE, Tableau.

Languages: English, Tamil (Native), Hindi, Malayalam, German.

Research Interests

Atmospheric rivers and moisture transport; Monsoon dynamics; Extreme precipitation and Impacts; Land–Atmosphere interactions; Numerical Weather Prediction; Climate Change; SDGs.

Selected Publications

2026 [pandidurai et al. \(2026\)](#), International Journal of Climatology

2025 [Pandidurai et al. \(2025\)](#), Climate Dynamics

2025 [Saini et al. \(2025\)](#), Monthly Weather Review

2024 [Mallick et al. \(2024\)](#), Natural Hazards

Leadership & Engagement

Design Coordinator for IAHS 2025; Student Coordinator for Indo-German School (SLESS-2025); General Secretary (2018-19) for MCC Physics Society leading 150+ students. Regular conference presenter (EGU 2026, IAHS 2025, RWC 2024, INTROMET 2025, CORDEX 2023,).

Training & Memberships

Training: ML for Earth System (ICTS-TIFR), Data Assimilation (ICTS-B), Climate Models (IISc), Google Data Analytics professional certificate. **Memberships:** EGU, IAHS, SAMA and YHS India.

Hobbies

Travel Sketching, Running, Race walking, Photography and Digital Design